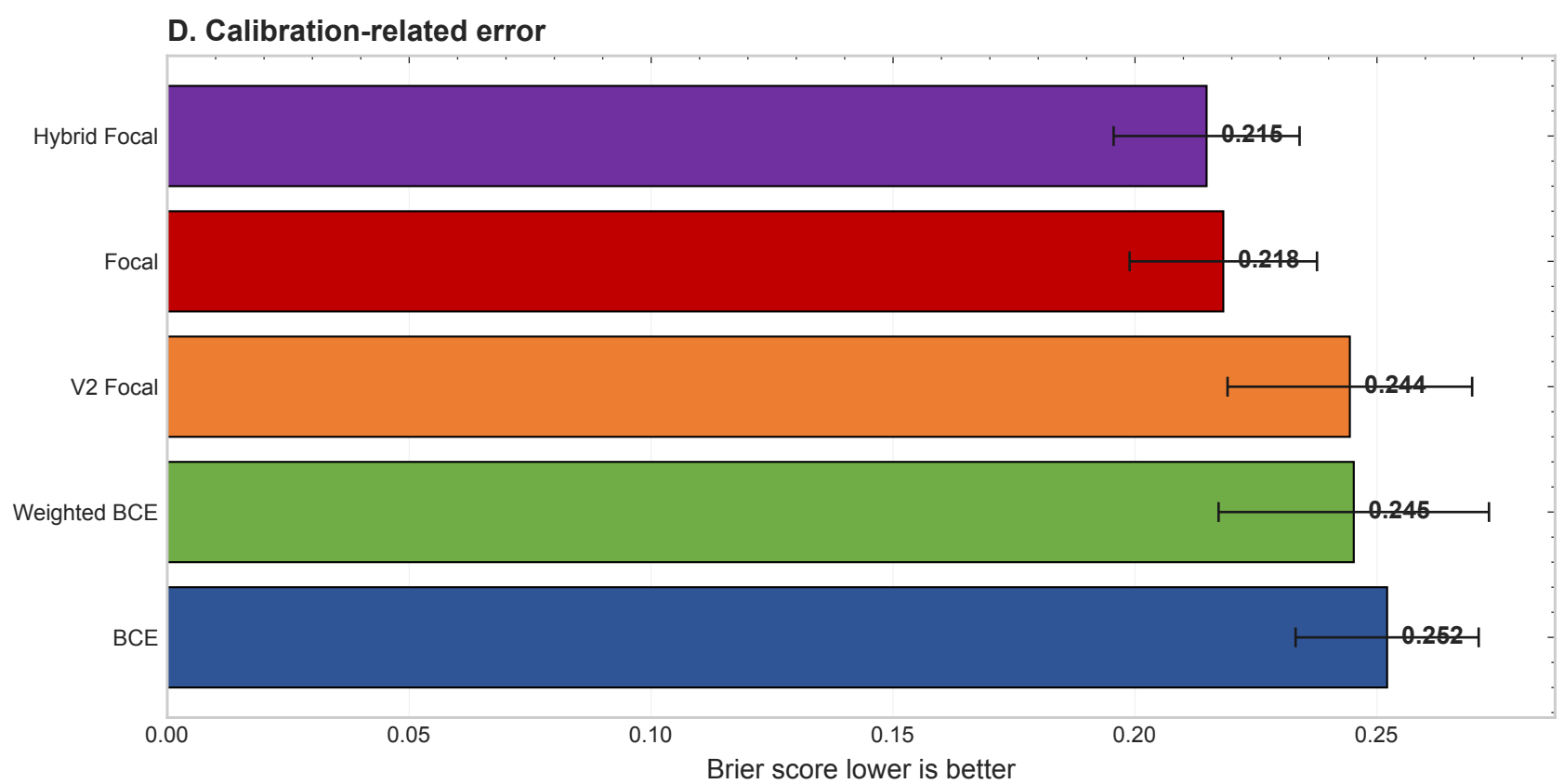
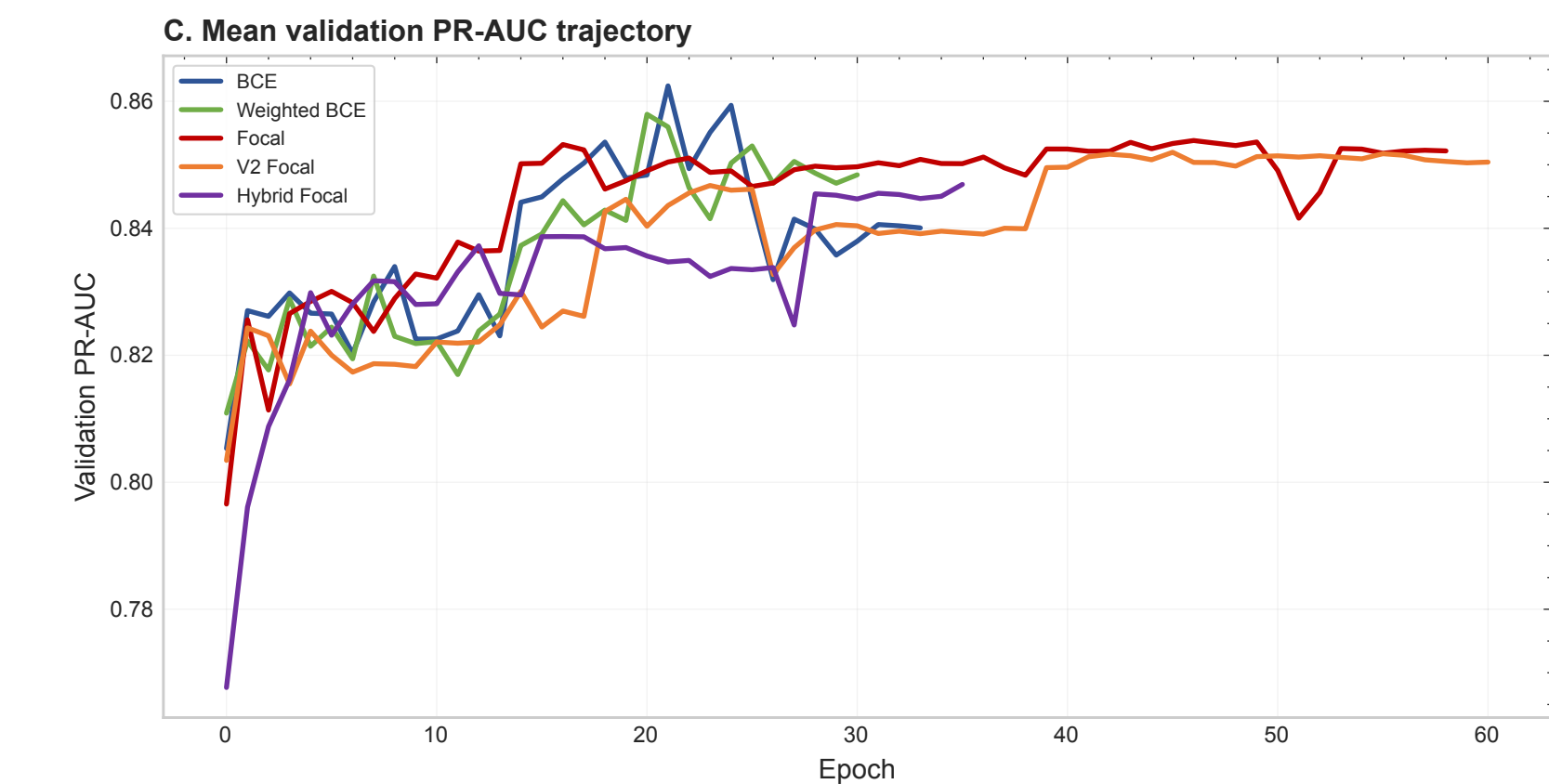
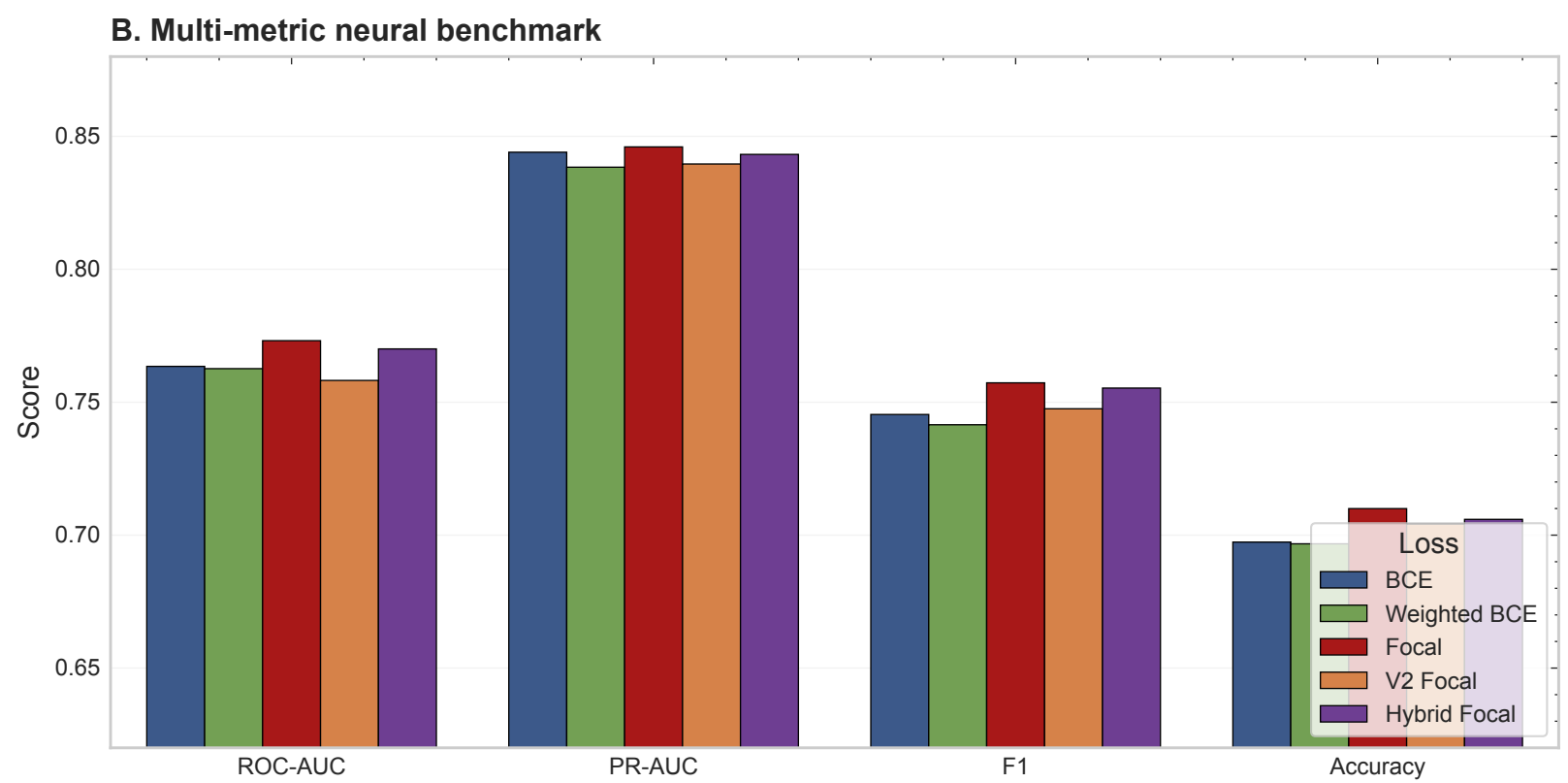
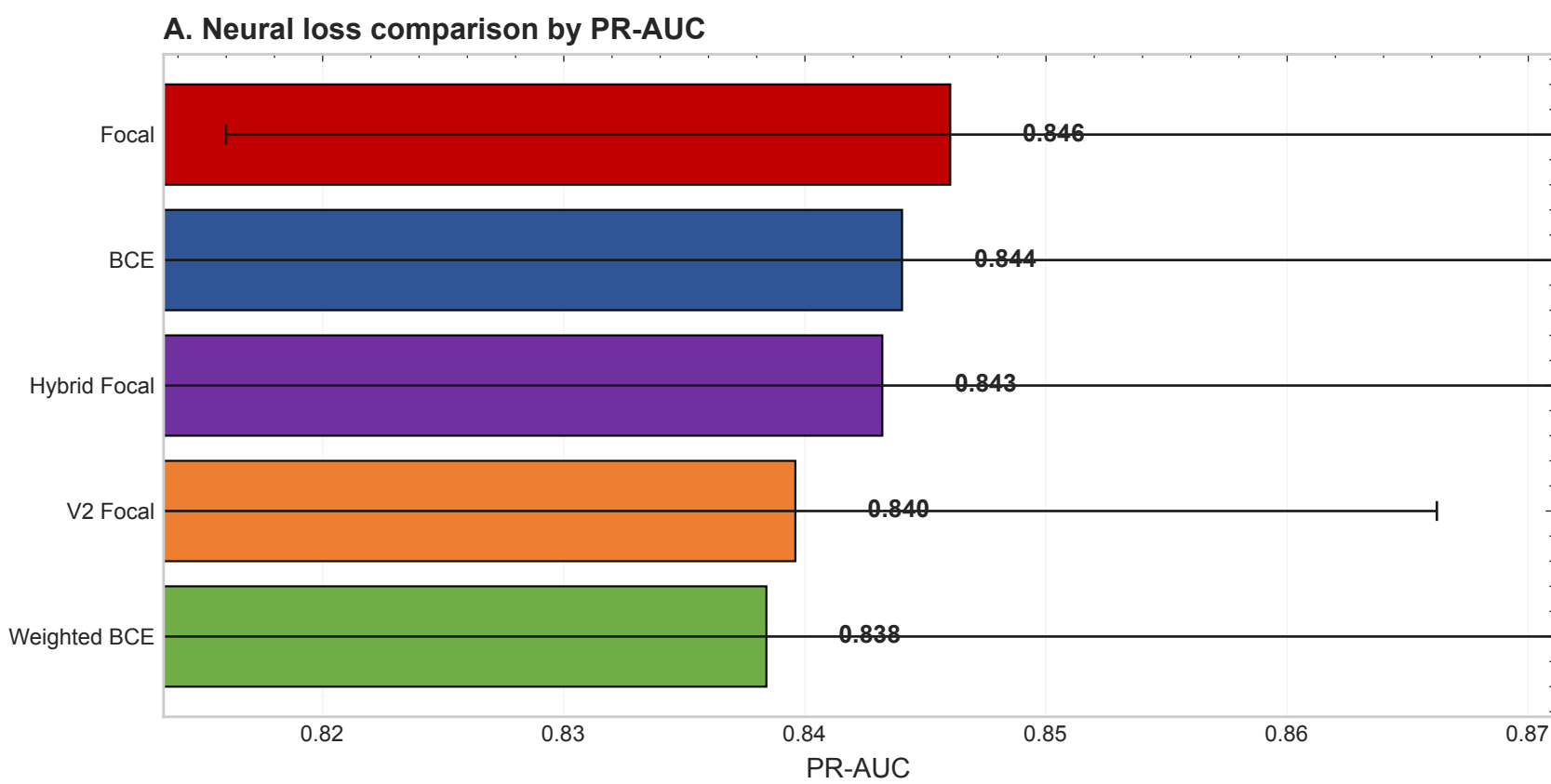


Figure 5. Neural pair scorer benchmark across loss functions



E. Neural model performance summary

Model	ROC-AUC	PR-AUC	F1	Accuracy	Precision	Recall	Brier	Median best epoch
BCE	0.763	0.844	0.745	0.697	0.767	0.726	0.252	8
Weighted BCE	0.763	0.838	0.742	0.697	0.777	0.712	0.245	3
Focal	0.773	0.846	0.757	0.710	0.774	0.741	0.218	10
V2 Focal	0.758	0.840	0.748	0.704	0.780	0.719	0.244	10
Hybrid Focal	0.770	0.843	0.755	0.706	0.771	0.741	0.215	12